

DO PHYSICS FOUNDATION MODELS LEARN GENERALIZABLE PHYSICS?

A BIAS-AWARE BENCHMARK ACROSS PHYSICAL REGIMES AND DISTRIBUTION SHIFTS

Mengdi Chu¹ Yang Liu¹ Ayan Biswas² Han-Wei Shen¹

¹The Ohio State University ²Los Alamos National Laboratory
 {chu.752, liu.12870, shen.94}@osu.edu ayan@lanl.gov

ABSTRACT

Recent physics foundation models claim general spatiotemporal forecasting ability, yet their evaluations often collapse performance into a single average score under a fixed training distribution. This makes it difficult to determine whether a model has learned generalizable physical dynamics or only performs well under particular settings. We construct a benchmark with 8 physical dynamics, 3 training-data mixtures, and 25 test regimes induced by dynamic-scale and initial-condition complexity shifts, covering in-distribution, distribution-shift, and out-of-distribution settings. We evaluate five physics foundation model architectures and four model variants per architecture (scratch and three pretrained sizes), resulting in 60,000 measurements. Our results show that current physics foundation models behave as conditional rather than universal generalists: their generality depends on the physical regime, temporal scale, initial-condition setting, pretraining, model size, and architecture. Improving the training data distribution only partially mitigates this limitation. Pretraining and scaling are also unable to reliably remove their ability biases. We argue that improving physics foundation models requires moving beyond scaling models or expanding data, toward learning mechanisms that better capture transferable physical knowledge across regimes, temporal scales, and distribution shifts. **PhysBiasBench** can be accessed at <https://huggingface.co/datasets/90879c/PhysBiasBench>.

1 INTRODUCTION

Foundation models are increasingly being extended from language and vision to physical forecasting and scientific simulation. Across learned physical simulators (Battaglia et al., 2016; Sanchez-Gonzalez et al., 2020; Pfaff et al., 2021), data-driven physical forecasting systems (Pathak et al., 2022; Lam et al., 2023; Bodnar et al., 2025), neural operators and neural PDE solvers (Lu et al., 2021; Li et al., 2021; Kovachki et al., 2023; Brandstetter et al., 2022), and PDE foundation models (Hao et al., 2024; Herde et al., 2024; McCabe et al., 2024), these models are often motivated by the hope that one trained model can be reused across physical regimes rather than rebuilt for each equation or setting. This motivates a central question: *do physics foundation models learn transferable physical knowledge, or do they mainly learn strong predictors for the distributions they have seen?* However, current evaluation protocols often make this question difficult to answer. Existing PDE benchmarks broaden the set of equations and tasks (Takamoto et al., 2022; Ohana et al., 2024; Köhler et al., 2024; Hassan et al., 2023), but model comparisons are often reduced to a single average score under one training distribution. Such averages are useful for leaderboard ranking, but they hide the conditional structure of physical generalization. For example, a model may perform well on some physical regimes while failing on others; it can perform well at a familiar temporal scale but fail under temporal extrapolation; it can benefit from pretraining in train-seen regimes while transferring harmful biases under distribution shift. Understanding these capability patterns and generalization boundaries is especially important in physical systems.

We therefore study *conditional generalization* in physics foundation models and separate three questions that are usually coupled. First, *is model capability uniformly general, or does it exhibit systematic capability biases across PDE families and distribution-shift types?* Second, *can changing the training-data distribution close the out-of-distribution gap, or does it mainly improve familiar regimes?* Third, *do model interventions such as pretraining and scaling fix these biases, or do they introduce their own conditional preferences?* Rather than asking only which model has the lowest error, we ask where each model generalizes, where it fails, and which factors explain the failure.

We design the benchmark to vary the main axes along which generality can break. We generate the PDE trajectories with the APEBench/Exponax toolkit (Köhler et al., 2024). The physical axis contains 8 PDE dynamics spanning reaction–diffusion, fluid-like, chaotic, and wave-like regimes. The data axis contains three training mixtures with different biases toward simple, balanced, or complex regimes. The test axis forms a 5×5 grid over dynamic-scale and initial-condition complexity, covering train-seen cells, interpolation-style distribution shifts, and extrapolative OOD regimes. The temporal axis evaluates both in-horizon prediction and longer-horizon rollout beyond the training rollout length. On top of these data axes, we evaluate five physics foundation model architectures: DPOT (Hao et al., 2024), GPhyT (Wiesner et al., 2025), MORPH (Rautela et al., 2025), MPP (McCabe et al., 2024), and Poseidon (Herde et al., 2024), with four variants per architecture, including scratch models and three pretrained sizes from small to large. Together, this design yields 60,000 aggregated measurements across physical dynamics, training mixtures, test regimes, model variants, and prediction horizons.

Our results show that current physics foundation models behave as conditional rather than universal generalists. Their performance depends strongly on the physical regime, temporal scale, initial-condition setting, training mixture, pretraining state, model size, and architecture. For example, under the Mix-balance setting, within-model normalized error is consistently low on Fisher–KPP and Gray–Scott, but rises sharply on Wave dynamics, reaching above $2\times$ the model-average error for several variants. Distribution shift further exposes this conditionality: dynamic-scale OOD regimes produce much larger ShiftDamage than initial-condition shifts, with the hardest cells reaching roughly $7\text{--}8\times$ the train-seen reference error. Increasing training complexity only partially mitigates the problem; moving from Mix-simple to Mix-complex reduces in-distribution raw error but increases normalized ShiftDamage across OOD groups. Model-side interventions are also conditional: under matched-size comparison, 37.5% of architecture–PDE pairs show negative pretraining transfer, and larger pretrained variants are worse than the small baseline in 25.0% of larger-model cells. These findings suggest that the limitation of current physics foundation models is not only insufficient data or model size, but limited ability to learn transferable physical structure across regimes, temporal scales, and distribution shifts. This paper makes three contributions:

2 RELATED WORK

The foundation-model paradigm is increasingly being extended from language and vision to scientific and physical domains, with the goal of moving from task-specific solvers toward reusable models across equations and regimes (Bommasani et al., 2021). Neural operators such as DeepONet (Lu et al., 2021) and the Fourier Neural Operator (Li et al., 2021) learn maps between function spaces, while physics-informed and scientific-ML methods use physical constraints to regularize learning (Raissi et al., 2019; Willard et al., 2022). Recent PDE foundation models extend this direction by pretraining transformer- or operator-style architectures across multiple equations and regimes. DPOT (Hao et al., 2024), Poseidon (Herde et al., 2024), GPhyT (Wiesner et al., 2025), MORPH (Rautela et al., 2025), and MPP (McCabe et al., 2024) all aim at broad PDE forecasting ability, but their evaluations are often summarized by average error under one training distribution.

An important ambiguity in learned physical models is whether low error reflects physical reasoning or high-quality pattern matching. Low forecasting error does not necessarily imply transferable physical understanding. Neural networks can exploit shortcuts, spurious correlations, or distribution-specific regularities while still achieving high benchmark accuracy (Shah et al., 2020; Geirhos et al., 2020). This concern is important for PDE forecasting, where a model may fit familiar regimes well but fail when the temporal scale, initial-condition complexity, or rollout horizon changes. Existing evaluations often make this hard to see because they report aggregate error under a limited set of training and test conditions.

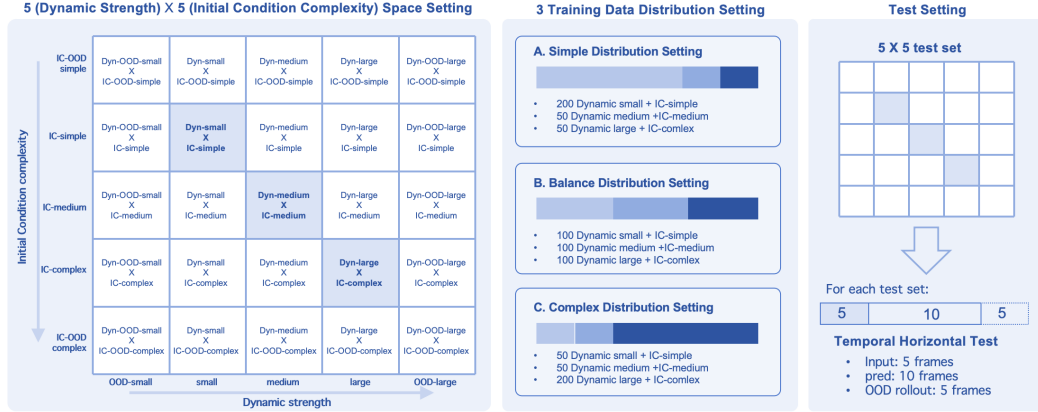


Figure 1: Overview of the benchmark protocol. The evaluation space is organized as a 5×5 grid over dynamic-scale setting and initial-condition complexity. The three diagonal cells correspond to train-seen regimes, while the remaining cells test compositional in-range shifts, dynamic-scale OOD shifts, IC-complexity OOD shifts, and joint OOD shifts. We construct three training mixtures with different biases toward simple, balanced, or complex regimes, while evaluating every model on the same full 5×5 test grid. For each test case, models are given 5 input frames, predict 10 in-horizon frames, and are further evaluated on 5 additional OOD rollout frames.

Scientific-ML benchmarks such as PDEBench (Takamoto et al., 2022), The Well (Ohana et al., 2024), APEBench (Köhler et al., 2024), BubbleML (Hassan et al., 2023), and CFDBench (Luo et al., 2023) have expanded the empirical coverage of learned physical models. A related line of work asks not only how accurate a model is, but what capability patterns and biases are hidden by aggregate scores. Domain-generalization and group-robustness benchmarks such as WILDS (Koh et al., 2021) and DomainBed (Gulrajani & Lopez-Paz, 2021) expose group-dependent failures under distribution shift, while model-bias studies show that high-level performance can reflect shortcut cues or pretrained priors rather than input-grounded reasoning (Geirhos et al., 2020; Shah et al., 2020; Vo et al., 2025). Recent physical-understanding benchmarks for vision-language models, such as PhysBench (Chow et al., 2025) and QuantiPhy (Li et al., 2025), similarly evaluate whether general-purpose models possess reliable physical capabilities rather than plausible responses. However, existing physical foundation model evaluations rarely factorize the sources of capability. As a result, it remains difficult to tell whether a model’s apparent generality reflects transferable physical capability or strong performance under particular regimes. Therefore, our work connects these ideas in the setting of PDE foundation models. We evaluate apparent generality through controlled changes in physical dynamics, training mixture, test regime, prediction horizon, pretraining state, model size, and architecture.

3 BENCHMARK AND PROTOCOL DESIGN

We factorize model evaluation into physical dynamics, training distribution, test-regime shift, prediction horizon, pretraining state, model size, and architecture. This section defines these axes and the diagnostic metrics used in the rest of the paper.

3.1 FACTORIZED EVALUATION AXES

Each measurement is indexed by

$$E_H(m, v, p, r, c),$$

where m is the architecture, v is the model variant, p is the physical dynamics, r is the training mixture, and $c = (d, q)$ is the test cell. Here, d denotes the dynamic-scale setting and q denotes initial-condition complexity. H denotes the prediction horizon. Table 1 summarizes the role of each axis. The complete grid yields $5 \times 4 \times 8 \times 3 \times 25 \times 5 = 60,000$ aggregated measurements.

Table 1: Factorized evaluation axes. The benchmark includes variation of data distribution, physics dynamics, temporal horizon, and model factors separately so that average performance can be decomposed into interpretable capability biases.

Axis	Values	What it tests
Architecture m	DPOT, GPhyT, MORPH, MPP, Poseidon	Architecture-dependent inductive bias
Variant v	Scratch and pretrained S/M/L variants	Pretraining transfer and scaling behavior
Physical dynamics p	Fisher-KPP, Gray-Scott, Swift-Hohenberg, Burgers, Kolmogorov, Kuramoto-Sivashinsky, Decay, Wave	PDE-family and physics-regime bias
Training mixture r	Mix-simple, Mix-balance, Mix-complex	Whether data coverage changes the bias pattern
Test cell (d, q)	5×5 dynamic-scale $\times$ IC-complexity grid	Compositional, temporal-ODD, IC-ODD, and joint-ODD generalization
Horizon H	1-step, 5-step, 10-step, OOD rollout, overall steps	Short-term accuracy and longer-rollout stability

3.2 TRAINING MIXTURES

The training distribution is treated as an experimental treatment rather than a fixed background choice. We construct three mixtures with controlled proportions over the three train-seen diagonal regimes (Figure 1). Mix-simple overweights the simple diagonal regime, Mix-balance allocates data evenly across the three diagonal regimes, and Mix-complex overweights the complex diagonal regime. The total training-set size, model architecture, evaluation protocol, and test cells are held fixed across mixtures. This design lets us ask whether richer or more complex training data changes the capability profile, or only changes average error.

3.3 TEST REGIMES

All trajectories are generated procedurally using APEBench and its Exponax reference simulator (Köhler et al., 2024). For each PDE and initial-condition setting, we first simulate a dense trajectory of approximately 100 frames. The IC axis is controlled through the APEBench initial-condition generator, whose parameters change the spatial complexity of the starting field. The dynamic-scale axis is constructed after simulation by temporally subsampling the dense trajectory. Consecutive frames produce the smallest apparent dynamics, while larger frame strides produce larger state changes between prediction steps. Thus, the dynamic labels describe the effective temporal change seen by the forecasting model: *small*, *medium*, and *large* are in-range stride levels, while *OOD-small* and *OOD-large* are out-of-range stride levels on the slower and faster sides.

For each PDE family, test distributions form a 5×5 grid over dynamic-scale setting and IC complexity:

$$\begin{aligned} \text{Dynamic} &\in \{\text{OOD-small, small, medium, large, OOD-large}\}, \\ \text{IC} &\in \{\text{OOD-simple, simple, medium, complex, OOD-complex}\}. \end{aligned}$$

The three diagonal cells (small, simple), (medium, medium), and (large, complex) are train-seen regimes. The other cells are grouped into four interpretable shift types: *Compositional ID* for unseen in-range combinations, *Dynamic OOD* for out-of-range dynamic scale, *IC OOD* for out-of-range IC complexity, and *Joint OOD* when both axes are out of range. Figure 1 shows the layout. Moving horizontally changes the dynamic-scale setting while holding IC complexity fixed; moving vertically changes IC complexity while holding the dynamic-scale setting fixed; corners combine both shifts.

3.4 DIAGNOSTIC METRICS

Our base metric is per-sample-frame relative L_2 error, averaged over test trajectories and frames:

$$E_H = \frac{1}{|\mathcal{S}||H|} \sum_{i \in \mathcal{S}} \sum_{t \in H} \frac{\|\hat{u}_{i,t} - u_{i,t}\|_2}{\|u_{i,t}\|_2}. \quad (1)$$

Computing the relative error per sample and frame before aggregation avoids allowing high-energy trajectories, or frames with larger absolute field magnitude, to dominate the metric. Unless otherwise stated, non-rollout analyses use the 10-step in-horizon error. Long-horizon rollout analyses are stated explicitly. When reporting group-level uncertainty, we use paired 95% bootstrap confidence intervals obtained by resampling the 50 test trajectories. Average error gives only the overall ranking; it does not show which PDE families, shifts, horizons, or model choices drive the score. We

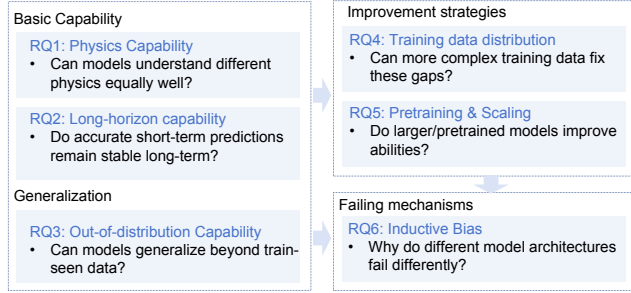
Table 2: Diagnostic metric suite. Unless otherwise stated, all quantities use the same prediction horizon when forming ratios. *Train-seen* denotes the three diagonal cells; *Compositional ID* denotes unseen in-range combinations; and *Dynamic/IC/Joint OOD* denote out-of-range shifts along one or both axes. For PretrainingGain, the scratch checkpoint is compared only with the size-matched finetuned M variant to avoid confounding pretraining with model size. $E_{1\text{-step}}$ denotes the first predicted-frame error, and E_{roll} denotes the explicitly stated rollout horizon error.

Metric	Definition	Interpretation
Base error	$E_H(m, v, p, r, c)$	Per-sample-frame relative L_2 error
PDEBias	$E(m, v, p) / \text{mean}_{p'} E(m, v, p')$	> 1 : model is relatively weak on PDE p
ShiftDamage	$E(m, v, p, c) / \text{mean}_{c' \in \mathcal{C}_{\text{train}}} E(m, v, p, c')$	> 1 : degradation relative to train-seen cells
RolloutAmplification	$E_{\text{roll}} / E_{1\text{-step}}$	Long-horizon growth from short-horizon error
PretrainingGain	$(E_{\text{scr}, M} - E_{\text{ft}, M}) / E_{\text{scr}, M}$	> 0 : matched-size pretraining improves accuracy
ModelSizeGain	$(E_{\text{ft}, S} - E_{\text{ft}, s}) / E_{\text{ft}, S}, s \in \{M, L\}$	> 0 : larger finetuned model improves over S

therefore define diagnostics that normalize error along the main axes of the benchmark: PDE family, test-cell shift, rollout horizon, pretraining state, and model size (Table 2).

4 RESULTS

We organize the results as a four-stage diagnosis. First, we evaluate basic capability across physical dynamics and rollout horizon (RQ1–RQ2). Second, we test whether that capability generalizes beyond train-seen dynamic and initial-condition regimes (RQ3). Third, we ask whether common improvement strategies—richer training mixtures, pretraining, and scaling—close the observed gaps (RQ4–RQ5). Finally, we summarize architecture-specific failure fingerprints to explain why models with similar average errors can fail in different ways (RQ6).



4.1 RQ1: DO PHYSICS FOUNDATION MODELS EXHIBIT UNIFORM CAPABILITY ACROSS PHYSICAL DYNAMICS?

We first test whether model capability is uniform across physical dynamics. We use Mix-balance, the three train-seen diagonal cells, and the 10-step horizon to isolate PDE-family effects from distribution-shift effects. Under this controlled setting, performance differences mainly reflect sensitivity to the physical dynamics rather than extrapolation.

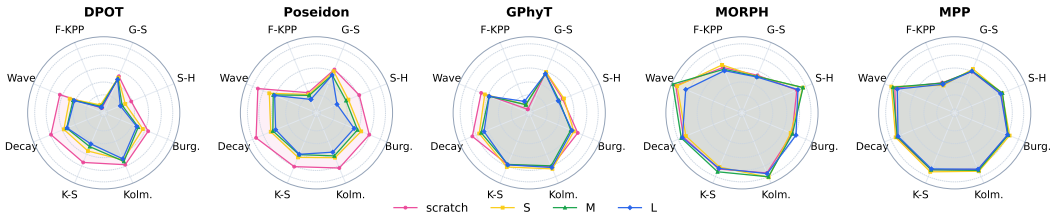


Figure 2: **Raw PDE performance.** Each plot corresponds to one model family, and each curve shows one model variant: scratch, S, M, and L. The eight axes denote PDE families, and the plotted value is the raw relative L_2 error averaged over the three train-seen cells and the 10 in-horizon predicted frames. Lower values indicate better absolute accuracy. The radial axis is shown on a log scale to make both low-error and high-error PDE regimes visible in the same plot.

Figure 2 shows that raw accuracy is highly PDE-dependent even under train-seen conditions. Fisher–KPP is the easiest regime for most variants: its median relative L_2 error across the 20 variants

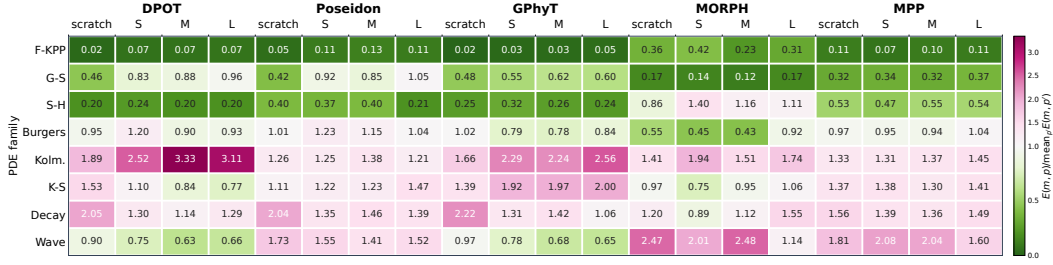


Figure 3: **PDEBias**. Rows correspond to PDE families and columns correspond to model variants, grouped by architecture and variant type. Each cell reports the within-model normalized error $E(m, p)/\text{mean}_{p'} E(m, p')$, computed using relative L_2 error averaged over the three train-seen cells and the 10 in-horizon predicted frames. Values below 1 indicate PDE families on which a model is relatively stronger than its own average, while values above 1 indicate relatively harder PDE families. Unlike the raw radar plot, this normalization removes each model’s overall error scale and highlights whether its capability is balanced across PDE families.

is 0.011, and it is the lowest-error PDE for DPOT, GPhyT, MPP, and Poseidon. Gray–Scott is also consistently low-error and is the easiest PDE for all MORPH variants. In contrast, the largest errors concentrate on higher-variation dynamics. Kolmogorov is the hardest PDE for most DPOT and GPhyT variants, while Wave dominates the failures of MORPH and MPP; the worst case is MORPH-M on Wave ($E = 1.33$). Thus, even on train-seen distributions, model performance can vary by one to two orders of magnitude across PDE families.

Raw error shows absolute performance, but it is not sufficient to characterize whether a model has a balanced capability profile across PDE families. A model with low overall error can still be disproportionately strong on some physical regimes and weak on others, while a less accurate model may perform more uniformly. Therefore, to evaluate physical-regime bias within each model, we compute a within-model normalized PDEBias:

$$\text{PDEBias}(m, p) = \frac{E(m, p)}{\text{mean}_{p'} E(m, p')},$$

where $E(m, p)$ is the relative L_2 error of model variant m on PDE family p , which compares each PDE against the model’s own average PDE error, separating a model’s PDE-specific preference from its overall accuracy level. Values below 1 indicate that the PDE is relatively easier for that model, while values above 1 indicate that it is relatively harder. As shown in Figure 3, the PDE dependence remains. Most variants are relatively strong on Fisher–KPP, Gray–Scott, and Swift–Hohenberg, while Kolmogorov, Kuramoto–Sivashinsky, Decay, and Wave frequently appear as above-average difficulty regimes. We can also see the bias pattern is model architecture dependent. MORPH shows a stronger relative weakness on Wave than DPOT and GPhyT, whereas DPOT and GPhyT show stronger relative difficulty on Kolmogorov dynamics. Scratch, S, M, and L variants within the same architecture often preserve similar column-wise trends, suggesting that pretraining and scaling modify the magnitude of PDE-specific errors but do not fully remove the underlying physical-regime preference. Together, the raw-error and normalized PDEBias results show that model capability varies strongly across physical dynamics and current physics foundation models exhibit systematic PDE-family-dependent capability bias even under train-seen conditions.

Finding 1. Physics foundation models exhibit systematic physical-regime bias. Their capability is relatively stronger on several reaction–diffusion and pattern-forming regimes, but weaker on high-variation or wave-like regimes such as Kolmogorov, Kuramoto–Sivashinsky, Decay, and Wave. Pretraining and scaling change the magnitude of errors, but do not fully remove these PDE-specific capability profiles.

4.2 RQ2: DO PHYSICS FOUNDATION MODELS MAINTAIN CAPABILITY ACROSS PREDICTION HORIZONS?

We next examine horizon-dependent forecasting capability. Under the same controlled setting as RQ1, we compare raw frame-wise error with normalized error amplification across prediction horizons, which separates immediate accuracy from temporal stability, for example, models with similar short-horizon errors may accumulate errors at very different rates, and models with weaker early predictions may be more stable over longer rollouts. Let

$$\text{RolloutAmplification}_{10}(m, p) = \frac{E_{10\text{-step}}(m, p)}{E_{1\text{-step}}(m, p)},$$

where $E_{1\text{-step}}$ is the first predicted-frame error and $E_{10\text{-step}}$ is the average error over the 10 in-horizon predicted frames. This ratio measures how strongly a model’s own initial error is amplified over the standard prediction horizon.

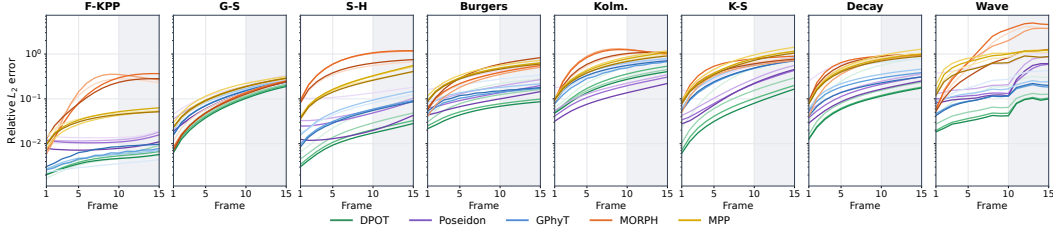


Figure 4: **PDE-wise rollout error growth.** Each subplot is one PDE family. Curves show the frame-wise relative L_2 error of all model variants, with colors grouped by architecture and shade indicating scratch/S/M/L variants. The gray region marks frames beyond the 10-step training horizon. Errors generally grow with prediction time, but the growth pattern depends strongly on both PDE family and model variant.

	DPOT				Poseidon				GPhyT				MORPH				MPP			
	scratch	S	M	L	scratch	S	M	L	scratch	S	M	L	scratch	S	M	L	scratch	S	M	L
F-KPP	2.0x	1.6x	1.9x	1.8x	0.9x	1.0x	0.9x	0.9x	1.3x	1.7x	1.7x	2.0x	20.4x	31.5x	17.7x	8.3x	2.5x	3.6x	3.4x	3.2x
G-S	7.3x	7.4x	7.3x	7.6x	2.6x	2.9x	3.8x	3.8x	4.3x	4.7x	5.1x	5.1x	6.1x	7.5x	7.6x	8.5x	4.5x	4.3x	4.4x	4.4x
S-H	4.2x	3.5x	3.4x	3.3x	1.1x	1.3x	1.3x	1.2x	2.7x	3.4x	3.5x	3.7x	4.3x	6.4x	6.7x	4.7x	3.9x	5.0x	4.7x	4.1x
Burgers	3.5x	2.6x	2.1x	2.1x	1.8x	1.6x	1.6x	1.6x	2.5x	2.2x	2.2x	2.1x	4.0x	4.8x	3.6x	5.7x	3.3x	3.0x	3.3x	3.3x
Kolm.	4.1x	3.5x	3.7x	3.2x	2.0x	2.5x	2.8x	2.6x	3.7x	4.5x	3.8x	4.1x	5.7x	9.3x	8.9x	7.2x	4.7x	4.6x	4.6x	4.2x
K-S	9.4x	7.2x	6.9x	6.3x	2.1x	3.1x	3.1x	3.6x	4.4x	5.4x	4.7x	5.0x	5.0x	6.7x	5.8x	4.9x	5.4x	5.4x	5.1x	4.9x
Decay	9.2x	5.4x	5.2x	5.0x	2.3x	2.9x	3.2x	3.3x	3.9x	3.8x	3.4x	3.4x	5.4x	9.7x	6.7x	6.1x	5.9x	6.5x	5.7x	5.4x
Wave	3.0x	1.9x	1.9x	1.8x	1.3x	1.3x	1.2x	1.3x	2.6x	1.5x	1.9x	2.1x	16.6x	16.3x	22.1x	7.5x	4.7x	3.8x	4.7x	3.7x

Figure 5: **10-step in-horizon error amplification by model variant and PDE family.** Rows correspond to PDE families and columns correspond to model variants, grouped by architecture and variant type. Each cell reports $E_{10\text{-step}}/E_{1\text{-step}}$, where $E_{1\text{-step}}$ is the first predicted-frame error and $E_{10\text{-step}}$ is the average error over the 10 in-horizon predicted frames. Larger values indicate stronger error accumulation within the standard prediction horizon.

Figure 4 shows systematic but uneven temporal degradation. Errors usually increase with prediction time, especially after the gray OOD rollout region begins, but the growth curves differ substantially across architectures and PDE families. Some variants grow smoothly, whereas others show sharp late-stage amplification. This indicates that horizon-dependent capability has two components: the absolute error reached at later frames and the rate at which early errors are amplified.

Figure 5 quantifies how much each model’s first-frame error is amplified within the standard prediction horizon. Error amplification is strongly architecture- and PDE-dependent. Poseidon has the lowest mean amplification ($2.09\times$) despite having the largest mean first-frame error among the five architectures (0.072). In contrast, DPOT has the lowest mean first-frame error (0.019) but a larger amplification factor ($4.34\times$). MORPH illustrates the separation between short- and long-horizon behavior: its first-frame error is competitive on several PDEs, including Gray–Scott, Burgers, and

Wave, where it outperforms Poseidon and MPP on immediate prediction. However, these early advantages often do not persist. MORPH has the largest mean amplification ($9.11\times$), with extreme cells such as MORPH-S on Fisher–KPP ($31.5\times$) and MORPH-M on Wave ($22.1\times$). Across all 160 model–PDE test, first-frame error is strongly correlated with raw 10-step error (Spearman $\rho = 0.86$), but almost uncorrelated with 10-step amplification (Spearman $\rho = 0.04$). Therefore, early-frame accuracy and temporal stability capture different aspects of forecasting capability.

Finding 2. Physics foundation models exhibit horizon-dependent capability bias. Competitive early-frame accuracy does not necessarily translate into stable longer-horizon behavior: some models amplify errors rapidly, while others maintain lower growth despite weaker short-term accuracy. Current models therefore learn temporal dynamics unevenly across horizons rather than stable forecasting behavior that transfers across time.

4.3 RQ3: DO PHYSICS FOUNDATION MODELS GENERALIZE BEYOND TRAIN-SEEN DISTRIBUTIONS?

After analyzing physical-regime bias and horizon-dependent behavior, we ask how models generalize when the test condition moves away from the train-seen cells. Here, distribution shift is not a binary ID/OOD label. The 5×5 grid separates two physically meaningful directions of shift: dynamic scale and initial-condition complexity. This lets us ask which direction creates the larger generalization gap, and whether the answer depends on the PDE family. We measure this gap with ShiftDamage, which normalizes each shifted cell by the same model’s train-seen performance on the same PDE. For each model variant and PDE family, we define

$$\text{ShiftDamage}(m, p, c) = \frac{E(m, p, c)}{\text{mean}_{c' \in \mathcal{C}_{\text{train}}} E(m, p, c')},$$

where c is a test cell and $\mathcal{C}_{\text{train}}$ contains the three train-seen diagonal cells. ShiftDamage measures relative degradation from a model’s own train-seen baseline. A value near 1 means that the shifted cell is no harder than the train-seen reference for that model–PDE pair, while values above 1 indicate a robustness gap under the corresponding physical shift.

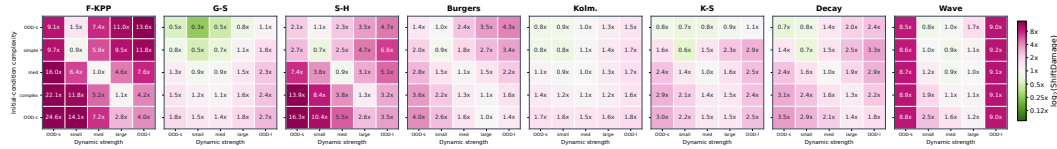


Figure 6: **PDE-wise 5×5 ShiftDamage grids under Mix-balance.** Each panel corresponds to one PDE family, computed at the 10-step horizon. Columns increase dynamic strength from OOD-small to OOD-large, and rows increase initial-condition complexity from OOD-simple to OOD-complex. Each cell reports ShiftDamage averaged over all evaluated model variants, with each variant normalized by its own train-seen diagonal baseline. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.

Figure 6 shows that distribution shift is organized by the direction of the physical change. Dynamic-scale shifts produce the clearest relative error growth. For Wave, the two Dynamic-OOD columns remain around $9\times$ across most IC settings, while the in-range dynamic cells stay much closer to the train-seen baseline. Fisher–KPP and Swift–Hohenberg show a similar sensitivity to dynamic-scale extrapolation. In contrast, Gray–Scott, Kolmogorov, and Kuramoto–Sivashinsky contain more near-baseline or reduced-error cells, especially around simpler IC settings. Thus, moving away from the train-seen diagonal does not have a single effect: depending on the PDE and shift direction, error can increase sharply, remain stable, or even decrease.

Appendix 27–31 provide the model-resolved view of this pattern. The broad PDE-dependent layout remains visible across architectures, but the contrast between train-seen and shifted cells changes with pretraining state and size. Larger or pretrained variants often reduce error on train-seen cells more than on shifted cells, which makes the normalized robustness gap more visible. Across Dynamic-OOD and Joint-OOD groups, mean ShiftDamage increases from $4.22\times$ for S variants

to $6.28\times$ for L variants, while scratch variants average $3.24\times$. Because this metric is normalized by each variant’s own train-seen baseline, the increase indicates uneven transfer of train-seen gains to shifted regimes. The group-level summary in Figure 26 gives the same message: median ShiftDamage is higher for Dynamic-ODD ($1.9\times$) and Joint-ODD ($1.6\times$) than for IC-ODD ($1.3\times$), and their much larger means ($4.8\times$ for both groups).

Finding 3. Physics foundation models do not generalize consistently beyond train-seen distributions. Their robustness boundary is structured by both PDE family and shift direction. Dynamic-scale and joint shifts cause the largest relative error growth, especially for Fisher–KPP, Swift–Hohenberg, and Wave, whereas many IC-only shifts remain close to the train-seen baseline or even reduce relative error.

4.4 RQ4: HOW DOES THE TRAINING DISTRIBUTION RESHAPE MODEL CAPABILITY?

The previous sections show that model capability depends on physical regime, prediction horizon, and test shift. We next ask how much of this conditional behavior can be changed by the training distribution. The three mixtures have the same total scale but allocate data differently across simple, balanced, and complex regimes. This separates two effects: whether a more complex mixture lowers absolute error, and whether those gains carry over to shifted test cells. We use the same normalized ShiftDamage as in RQ3, but compute the train-seen baseline separately for each training mixture:

$$\text{ShiftDamage}(t, m, p, c) = \frac{E(t, m, p, c)}{\text{mean}_{c' \in \mathcal{C}_{\text{train}}} E(t, m, p, c')},$$

where t is the training mixture, m is the model variant, p is the PDE family, and c is a test cell. This normalization asks whether a mixture improves unseen cells relative to the competence it achieves on its own train-seen cells.

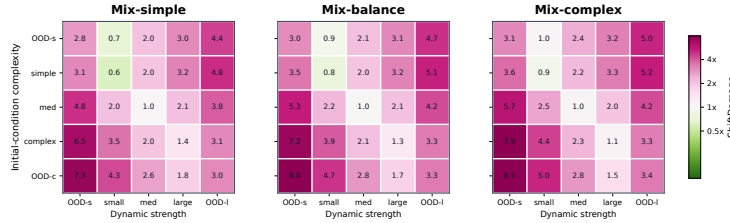


Figure 7: **Training-mixture effect on the 5×5 ShiftDamage grid.** Each panel corresponds to one training mixture. Rows vary initial-condition complexity and columns vary dynamic scale. Each cell reports mean normalized ShiftDamage averaged over model variants and PDE families, using the 10-step relative L_2 error. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to white, values below $1\times$ are green, and values above $1\times$ are red.

Table 3: **Raw accuracy and normalized robustness by training mixture.** Values are averaged over model variants, PDE families, and cells in each shift group. Raw error measures absolute relative L_2 error, while ShiftDamage normalizes each mixture by its own train-seen diagonal baseline.

Shift group	Raw error			ShiftDamage		
	Mix-simple	Mix-balance	Mix-complex	Mix-simple	Mix-balance	Mix-complex
Train-seen	0.240	0.225	0.224	1.00	1.00	1.00
Comp. ID	0.298	0.282	0.283	2.46	2.57	2.77
Dyn-ODD	0.436	0.428	0.424	4.33	4.75	4.98
IC-ODD	0.304	0.287	0.288	2.40	2.55	2.66
Joint-ODD	0.429	0.419	0.416	4.36	4.75	4.96

Table 3 separates absolute accuracy from relative robustness. In raw error, Mix-complex improves over Mix-simple in every shift group, including train-seen cells ($0.240 \rightarrow 0.224$), IC-ODD cells ($0.304 \rightarrow 0.288$), and Joint-ODD cells ($0.429 \rightarrow 0.416$). Thus, more complex training data does

increase average predictive capability. The improvement, however, is spatially selective. Figure 7 shows that Mix-complex improves high-dynamic cells most clearly relative to Mix-balance, including Dynamic-large $\times$ IC-complex (9.5% lower error), Dynamic-large $\times$ IC-OOD-complex (8.7% lower), and Dynamic-OOD-large $\times$ IC-complex (5.8% lower). The same mixture increases error in several small-dynamic cells, including Dynamic-small $\times$ IC-simple (7.6% higher error) and Dynamic-small $\times$ IC-OOD-simple (8.0% higher). Training data therefore redistributes capability across the test grid rather than improving every region together.

The ShiftDamage columns show that these raw gains do not close the relative robustness gap. Mix-complex has larger normalized damage than Mix-simple in every non-train group: 2.46 $\rightarrow$ 2.77 on compositional cells, 4.33 $\rightarrow$ 4.98 on Dynamic-OOD, 2.40 $\rightarrow$ 2.66 on IC-OOD, and 4.36 $\rightarrow$ 4.96 on Joint-OOD. Since the baseline is recomputed within each mixture, this means that train-seen gains transfer unevenly to shifted regimes. Dynamic-OOD and Joint-OOD remain the largest gaps under all three mixtures, so changing the training distribution reshapes model strengths but does not remove the conditional generalization behavior observed in RQ1–RQ3.

Finding 4. More complex training data shifts model strengths across regimes, but does not consistently improve generalization. It improves absolute accuracy in selected regions, especially large-dynamic regimes, but the gains are uneven and do not carry over across physical shifts. The persistent Dynamic-OOD and Joint-OOD gaps show that more complex training data alone is insufficient to produce robust OOD capability across the grid.

4.5 RQ5: HOW DO PRETRAINING AND SCALING RESHAPE MODEL CAPABILITY?

After studying data-side changes in RQ4, we now examine two model-side interventions: pretraining and scaling. The goal is not only to test whether they reduce error on average, but to ask whether the resulting gains are stable across physical regimes and test conditions. We therefore use controlled comparisons: pretraining is evaluated at matched size by comparing scratch- M and pretrained- M , while scaling is evaluated within pretrained variants using S as the baseline. Here, we define use PretrainingGain and ModelSizeGain:

$$\text{PretrainingGain}(a, p) = \frac{E_{\text{scratch}, M}(a, p) - E_{\text{ft}, M}(a, p)}{E_{\text{scratch}, M}(a, p)} \times 100.$$

$E(a, p)$ is the relative L_2 error for architecture a and PDE family p , averaged over the full grid. Positive values indicate beneficial transfer, while negative values indicate negative transfer. Scaling is evaluated only among pretrained variants by using the S model as the within-architecture baseline:

$$\text{ModelSizeGain}(a, p, s) = \frac{E_{\text{ft}}(a, p, S) - E_{\text{ft}}(a, p, s)}{E_{\text{ft}}(a, p, S)} \times 100, \quad s \in \{S, M, L\}.$$

By definition, the S column is zero; positive M/L values mean that a larger model helps, while negative values mean inverse scaling.

Figure 8 shows that pretraining reshapes capability in a regime-dependent way. It improves some architecture–PDE pairs, but does not produce a consistently better model. Across the 5×8 architecture–PDE cells, 37.5% show negative transfer and the mean gain is slightly negative (−2.3%). The effect is strongly conditional: Poseidon-M improves Decay by +38% but degrades Fisher–KPP by −199%, while GPhyT-M degrades Fisher–KPP by −64% but improves Decay by +20%. Thus, pretraining appears to transfer regime-specific preferences rather than a universally useful physics prior.

Figure 9 shows that scaling has the same selective character. Larger models often help: L is the best size in 23 of the 40 architecture–PDE pairs. However, scaling is not monotonic. Only 45.0% of architecture–PDE pairs improve monotonically from S to M to L , and 25.0% of M/L cells are worse than the S baseline. For example, MORPH-L improves Wave by +34% and Swift–Hohenberg by +38%, but MORPH-M is worse than S on Wave (−50%), and Poseidon-M/L are much worse than S on Fisher–KPP (−173% and −161%). Scaling therefore reallocates capability across regimes rather than improving all regimes together.

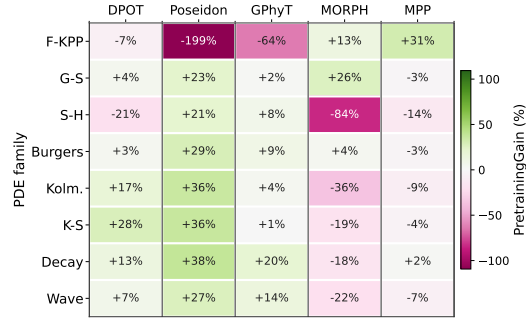


Figure 8: **Matched-size PretrainingGain under Mix-balance.** Rows correspond to PDE families and columns correspond to architectures. Each cell reports the percentage error reduction of pretrained- M relative to scratch- M for the same architecture, averaged over the full 25-cell grid at the 10-step horizon. Green indicates beneficial transfer and pink indicates negative transfer.

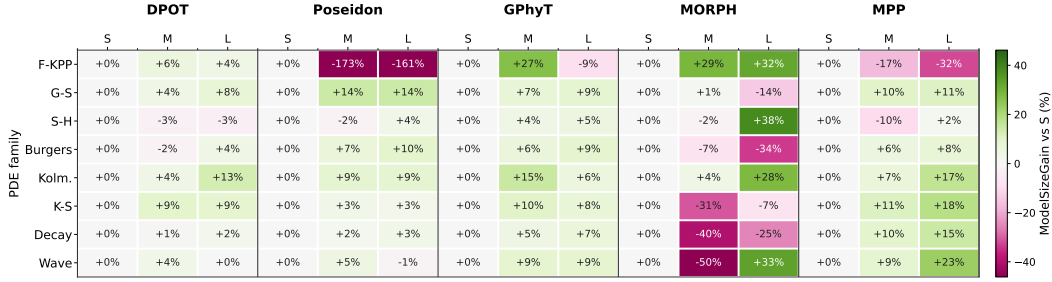


Figure 9: **ModelSizeGain under Mix-balance.** Rows correspond to PDE families and columns correspond to model variants, grouped by architecture and finetuned size. Each cell reports the percentage error reduction relative to the S finetuned model of the same architecture and PDE. The S column is therefore zero. Green indicates that the larger model improves over S ; pink indicates inverse scaling.

Finding 5. Pretraining and scaling produce selective capability gains. Pretraining is not universally beneficial: even under a matched-size comparison, 37.5% of architecture-PDE pairs show negative transfer. Scaling is also non-monotonic. Although the large variant is best in 23 of 40 pairs, 25.0% of larger-model cells are worse than the small baseline.

4.6 RQ6: DO PHYSICS FOUNDATION MODELS FAIL THROUGH DIFFERENT MECHANISMS?

The previous RQs show that failures arise along several axes: physical-regime bias, distribution shift, rollout stability, pretraining transfer, and scaling. We now ask whether architectures share the same weakness profile, or whether each architecture fails through a different combination of mechanisms. To make these profiles comparable, we aggregate five diagnostics under Mix-balance: *PDE bias severity* as the mean absolute log PDEBias across PDE families, *OOD ShiftDamage* as the mean normalized degradation over non-train-seen cells, *RolloutAmplification* as $E_{\text{rollout_ood5}}/E_{\text{id01}}$ on the three train-seen cells, *negative transfer rate* from matched scratch- M versus pretrained- M , and *inverse scaling rate* from pretrained M/L versus pretrained S .

Figure 10 shows that architectures fail in qualitatively different ways. DPOT is most sensitive to distribution shift: it has the largest normalized OOD degradation ($8.66\times$), even though its PDE bias severity is the lowest among the five models. MORPH is dominated by temporal and scaling instability, with the largest rollout amplification ($22.8\times$) and the highest inverse-scaling rate (56.2%). MPP shows a different profile: it has the highest matched-size negative-transfer rate (75.0%), but its temporal amplification is much smaller than MORPH’s. Poseidon and GPhyT are more balanced in this fingerprint, but still have clear weaknesses; Poseidon shows scaling sensitivity, while GPhyT

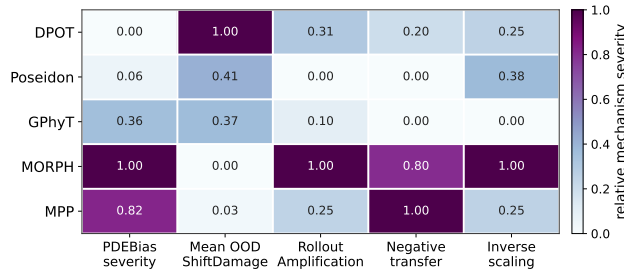


Figure 10: **Architecture-level failure-mechanism fingerprint.** Rows correspond to architectures and columns correspond to failure dimensions derived from the diagnostics above. Values are min–max normalized within each column, so darker cells indicate that an architecture is relatively more affected by that failure mode.

remains affected by PDE and rollout variation. These differences indicate that architecture matters not only through average accuracy, but through the type of failure it induces.

Finding 6. Architectures fail through different combinations of mechanisms rather than along a single accuracy axis. DPOT is most sensitive to distribution shift, MORPH to rollout and scaling instability, and MPP to pretraining negative transfer, while Poseidon and GPhyT show more balanced but still uneven profiles. Thus, improving physics foundation models requires addressing architecture-specific failure modes, not only lowering average error.

5 DISCUSSION

What current models are good for. Our results show that the learned competence of physical foundation model is conditional, which is different from the usual foundation-model promise of broad reuse across tasks and regimes (Bommasani et al., 2021). In our benchmark, the same model can be strong on one PDE family, one temporal horizon, or one shift direction, and weak on another. For practical use, this means that model choice should depend on the target regime rather than on a single averaged score. A model that is attractive for smooth dissipative dynamics may be a poor choice for wave-like or high-variation dynamics; a model with low short-horizon error may still be unsuitable for long rollouts. The useful object is therefore not just a model ranking, but a capability profile: where the model is accurate, where it degrades, and which failure mode is most relevant to the intended forecasting problem.

What data, pretraining, and scale buy. Much of foundation-model progress has been driven by larger models, larger datasets, and pretraining at scale (Kaplan et al., 2020; Hoffmann et al., 2022). Our results show that these levers also matter for physics forecasting, but their effects are selective. More complex training mixtures improve raw accuracy in targeted regions, especially cells with stronger dynamics, yet they leave large dynamic-scale and joint-shift gaps. Pretraining is also selective: it improves some architecture–PDE pairs, but can transfer regime preferences that hurt other pairs. Scaling behaves similarly. Larger variants are often better, but the gains are not monotonic and can reverse on particular PDE families or test cells. These patterns are consistent with a broader lesson from shortcut and simplicity-bias studies: high average accuracy does not guarantee that the model has learned the intended transferable structure (Geirhos et al., 2020; Shah et al., 2020).

The main bottleneck exposed by the benchmark is not only insufficient data or insufficient parameter count. The hard cases are organized: dynamic-scale shifts remain difficult, long-horizon errors grow differently across architectures, and pretraining or scaling can sharpen an existing regime preference instead of removing it. This suggests that the next improvement step should focus on how physical knowledge is represented and transferred. The architecture-level fingerprints also argue against a single universal fix: a model limited by rollout instability needs a different intervention from one limited by OOD sensitivity, negative transfer, or inverse scaling.

Limitations and next steps. This paper focuses on model behavior to show where current models succeed and fail, and how those failures change with the physical regime, data mixture, pretraining state, and model size. It does not yet explain what the models have learned internally. We see this as the next question. The error patterns should be tied back to physical quantities. More importantly, future analysis should ask whether these models can infer the governing regime from context. Given a few input frames, can the model recognize the relevant physical knowledge, and then use that knowledge outside the regimes it was trained on? Answering this would move beyond next-frame accuracy and toward whether physics foundation models learn reusable physical concepts.

6 CONCLUSION

This paper studies whether current physics foundation models learn broadly generalizable physical capability. Across 8 physical dynamics, 3 training mixtures, 25 test regimes, and 5 model architectures, we find that their capability is highly conditional. Models show systematic biases across PDE families, temporal horizons, distribution shifts, training mixtures, pretraining states, model sizes, and architectures. More complex data, pretraining, and scaling can improve selected regimes, but they do not reliably remove these biases or produce consistent OOD robustness. These results suggest that the next step for physics foundation models is not only to increase data scale or model size, but to learn physical representations that transfer more reliably across regimes, temporal scales, and distribution shifts.

REFERENCES

- Peter W. Battaglia, Razvan Pascanu, Matthew Lai, Danilo Jimenez Rezende, and Koray Kavukcuoglu. Interaction networks for learning about objects, relations and physics. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, Jayesh K. Gupta, Kit Tham-biratnam, Alexander T. Archibald, Chun-Chieh Wu, Elizabeth Heider, Max Welling, Richard E. Turner, and Paris Perdikaris. A foundation model for the earth system. *Nature*, 641:1180–1187, 2025. arXiv:2405.13063.
- Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- Johannes Brandstetter, Daniel E. Worrall, and Max Welling. Message passing neural PDE solvers. In *International Conference on Learning Representations (ICLR)*, 2022.
- Wei Chow, Jiageng Mao, Boyi Li, Daniel Seita, Vitor Campagnolo Guizilini, and Yue Wang. Phys-Bench: Benchmarking and enhancing vision-language models for physical world understanding. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2501.16411.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.
- Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Zhongkai Hao, Chang Su, Songming Liu, Julius Berner, Chengyang Ying, Hang Su, Anima Anandkumar, Jian Song, and Jun Zhu. DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 17616–17635. PMLR, 2024. arXiv:2403.03542.
- Sheikh Md Shakeel Hassan, Arthur Feeney, Akash Dhruv, Jihoon Kim, Youngjoon Suh, Jaiyoung Ryu, Yoonjin Won, and Aparna Chandramowlishwaran. BubbleML: A multiphase multiphysics dataset and benchmarks for machine learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

-
- Maximilian Herde, Bogdan Raonić, Tobias Rohner, Roger Käppli, Roberto Molinaro, Emmanuel de Bézenac, and Siddhartha Mishra. Poseidon: Efficient foundation models for PDEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.19101.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton Earnshaw, Imran Haque, Sara M. Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 5637–5664. PMLR, 2021.
- Felix Köhler, Simon Niedermayr, Rüdiger Westermann, and Nils Thuerey. APEBench: A benchmark for autoregressive neural emulators of PDEs. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. arXiv:2411.00180.
- Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Merose, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023.
- Puyin Li, Tiange Xiang, Ella Mao, Shirley Wei, Xinye Chen, Adnan Masood, Fei-Fei Li, and Ehsan Adeli. QuantiPhy: A quantitative benchmark evaluating physical reasoning abilities of vision-language models. *arXiv preprint arXiv:2512.19526*, 2025.
- Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *International Conference on Learning Representations (ICLR)*, 2021.
- Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- Yining Luo, Yingfa Chen, and Zhen Zhang. CFDBench: A large-scale benchmark for machine learning methods in fluid dynamics. *arXiv preprint arXiv:2310.05963*, 2023.
- Michael McCabe, Bruno Régalo-Saint Blancard, Liam Holden Parker, Ruben Ohana, Miles Cranmer, Alberto Bietti, Michael Eickenberg, Siavash Golkar, Géraud Krawezik, François Lanusse, Mariel Pettee, Tiberiu Tesileanu, Kyunghyun Cho, and Shirley Ho. Multiple physics pretraining for spatiotemporal surrogate models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2310.02994.
- Ruben Ohana, Michael McCabe, Lucas Meyer, Rudy Morel, Fruzsina J. Agocs, Miguel Beneitez, Marsha Berger, Blakesley Burkhart, Keaton Burns, Stuart B. Dalziel, Drummond B. Fielding, Daniel Fortunato, Jared A. Goldberg, Keiya Hirashima, Yan-Fei Jiang, Rich R. Kerswell, Suryanarayana Maddu, Jonah Miller, Payel Mukhopadhyay, Stefan S. Nixon, Jeff Shen, Romain Wateaux, Bruno Régalo-Saint Blancard, François Rozet, Liam H. Parker, Miles Cranmer, and

-
- Shirley Ho. The Well: A large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. arXiv:2412.00568.
- Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Anima Anandkumar. FourCastNet: A global data-driven high-resolution weather model using adaptive fourier neural operators. In *SC22: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2022.
- Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter W. Battaglia. Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations (ICLR)*, 2021.
- Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- Mahindra Singh Rautela, Alexander Most, Siddharth Mansingh, Bradley C. Love, Ayan Biswas, Diane Oyen, and Earl Lawrence. MORPH: Shape-agnostic PDE foundation models. *arXiv preprint arXiv:2509.21670*, 2025.
- Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. Learning to simulate complex physics with graph networks. In *International Conference on Machine Learning (ICML)*, 2020.
- Harshay Shah, Kaustav Tamuly, Aditi Raghunathan, Prateek Jain, and Praneeth Netrapalli. The pitfalls of simplicity bias in neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Dan MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.
- An Vo, Khai-Nguyen Nguyen, Mohammad Reza Taesiri, Vy Tuong Dang, Anh Totti Nguyen, and Daeyoung Kim. Vision language models are biased. *arXiv preprint arXiv:2505.23941*, 2025.
- Florian Wiesner, Matthias Wessling, and Stephen Baek. Towards a physics foundation model. *arXiv preprint arXiv:2509.13805*, 2025.
- Jared Willard, Xiaowei Jia, Shaoming Xu, Michael Steinbach, and Vipin Kumar. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Computing Surveys*, 55(4):1–37, 2022.

A APPENDIX: BENCHMARK DESIGN AND DATASET GENERATION

All benchmark trajectories are generated on two-dimensional periodic domains using the APEBench/Exponax pseudospectral solvers (Köhler et al., 2024). Unless otherwise stated, each stored group has shape $(N_{traj}, 100, C, 64, 64)$, where C is the number of physical channels. For every PDE we keep five initial-condition families ordered by complexity: IC-OOD-simple, IC-simple, IC-medium, IC-complex, and IC-OOD-complex. The training split contains 200 trajectories for each in-distribution family IC-simple, IC-medium, and IC-complex; validation and test contain 50 trajectories for each of the five families.

The dynamic-scale axis is constructed from these 100-frame trajectories by temporal subsampling. Each evaluation sequence contains 20 frames. The five dynamic settings use raw-frame strides 1, 2, 3, 4, 5, corresponding to Dynamic-OOD-small, Dynamic-small, Dynamic-medium, Dynamic-large, and Dynamic-OOD-large, respectively. For example, the stride-1 setting uses raw frames 0, 1, ..., 19, while the stride-5 setting uses 0, 5, ..., 95. This changes the effective temporal step seen by the forecasting model while keeping the PDE, spatial resolution, and IC family fixed. Figure panels below use one test trajectory per family, with rows ordered by IC complexity and columns showing ten uniformly sampled times.

A.1 GRAY-SCOTT

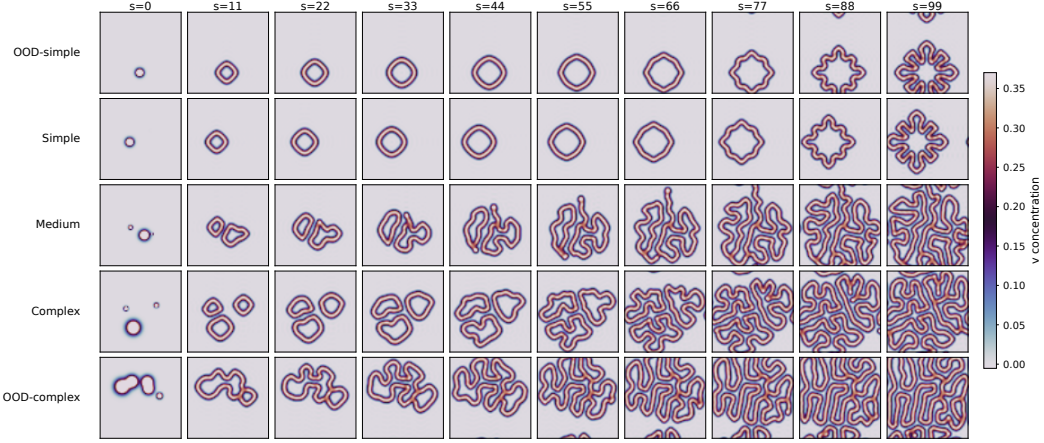


Figure 11: Gray-Scott examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

The Gray-Scott system models two reacting and diffusing chemical species u and v :

$$\partial_t u = D_u \Delta u + F(1 - u) - uv^2, \quad (2)$$

$$\partial_t v = D_v \Delta v - (F + k)v + uv^2. \quad (3)$$

We use the Exponax Gray-Scott stepper with periodic boundary conditions, domain extent $L = 1.35$, resolution 64×64 , solver step $\Delta t = 3.5$, and 14 internal solver steps per stored frame. Thus adjacent saved frames are separated by 49.0 time units. The parameters are $D_u = 2 \times 10^{-5}$, $D_v = 1 \times 10^{-5}$, feed rate $F = 0.044$, and kill rate $k = 0.06$. The two stored channels are the u and v concentrations; Figure 11 visualizes the v channel. Initial conditions are Gaussian-blob perturbations with $\sigma \in [0.002, 0.05]$, center fraction 0.5, and no additive noise. Complexity is controlled by the number of blobs: 1, 2, 3, 4, and 5 for OOD-simple, simple, medium, complex, and OOD-complex respectively.

A.2 WAVE

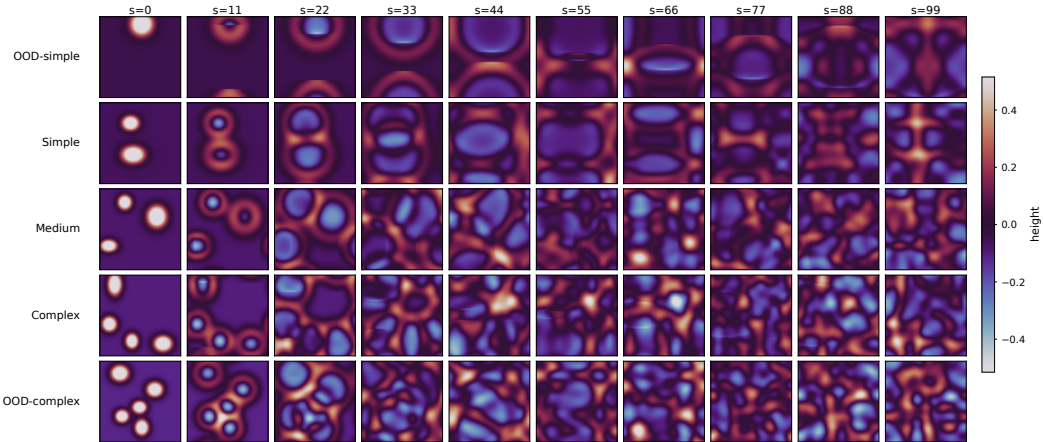


Figure 12: Wave-equation examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

The wave benchmark uses the second-order equation

$$\partial_{tt} h = c^2 \Delta h, \quad (4)$$

represented internally as the first-order system $\partial_t h = v$, $\partial_t v = c^2 \Delta h$. We store both height h and velocity v , and visualize h in Figure 12. The domain extent is $L = 1.0$, the resolution is 64×64 , the wave speed is $c = 1.0$, and saved frames use $\Delta t = 0.01$ with one internal step per frame. The height initial condition is generated from Gaussian blobs, while the initial velocity amplitude is zero. Blob centers are sampled in the coordinate range $[0.1, 0.9]$, variances are in $[0.002, 0.008]$, and the field is normalized to zero mean and unit max absolute value. IC complexity is controlled only by the number of blobs: 1, 2, 3, 4, and 5 from OOD-simple to OOD-complex.

A.3 FISHER–KPP

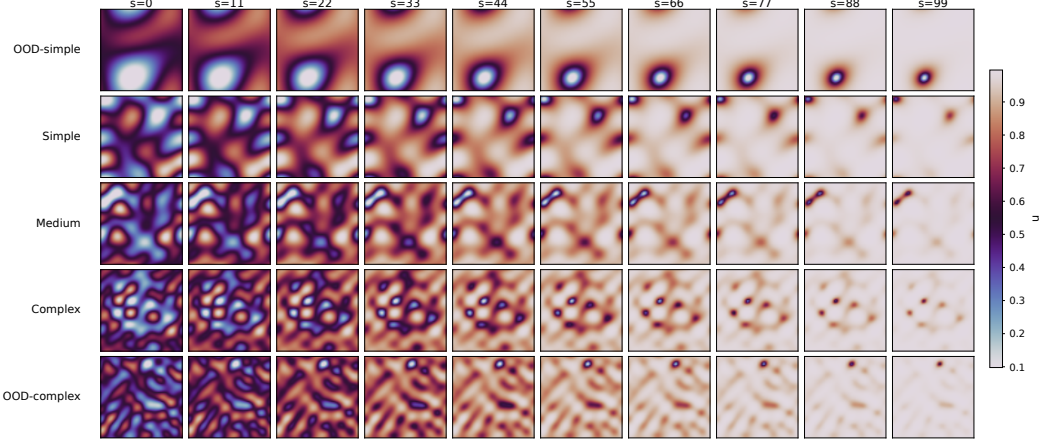


Figure 13: Fisher–KPP examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

The Fisher–KPP equation is

$$\partial_t u = D \Delta u + \rho u(1 - u), \quad (5)$$

with a single population channel u . We use $L = 4.0$, 64×64 resolution, solver step $\Delta t = 0.001$, two internal steps per stored frame, diffusivity $D = 0.005$, and reactivity $\rho = 20.0$. The saved frame spacing is therefore 0.002. Initial conditions are random truncated Fourier fields clamped and rescaled to the valid range $[0, 1]$. Complexity is controlled by the Fourier cutoff: 1, 2, 3, 4, and 5 for OOD-simple, simple, medium, complex, and OOD-complex respectively. Representative trajectories are shown in Figure 13.

A.4 BURGERS

The two-dimensional Burgers benchmark stores the vector velocity field $u = (u_x, u_y)$ and solves

$$\partial_t u + \frac{b}{2} \nabla \cdot (u \otimes u) = \nu \Delta u. \quad (6)$$

We use periodic boundary conditions, $L = 3.0$, 64×64 resolution, solver step $\Delta t = 0.02$, and two internal steps per stored frame. Parameters are diffusivity $\nu = 0.01$, convection scale $b = 2.0$, amplitude 0.5, ETDRK order 2, and Orszag 2/3 dealiasing. Figure 14 visualizes the velocity magnitude $\sqrt{u_x^2 + u_y^2}$. Initial conditions are random truncated Fourier fields; complexity is controlled by cutoff values 1, 2, 3, 4, and 5 from OOD-simple to OOD-complex.

A.5 SWIFT–HOHENBERG

The Swift–Hohenberg benchmark follows the Exponax reaction–diffusion form

$$\partial_t u = ru - (q_c + \Delta)^2 u + g(u), \quad g(u) = u^2 - u^3. \quad (7)$$

It is a single-channel pattern-formation system with periodic boundary conditions. We use $L = 30.0$, 64×64 resolution, saved-frame spacing $\Delta t = 0.05$, reactivity $r = 0.7$, critical number $q_c = 1.0$,

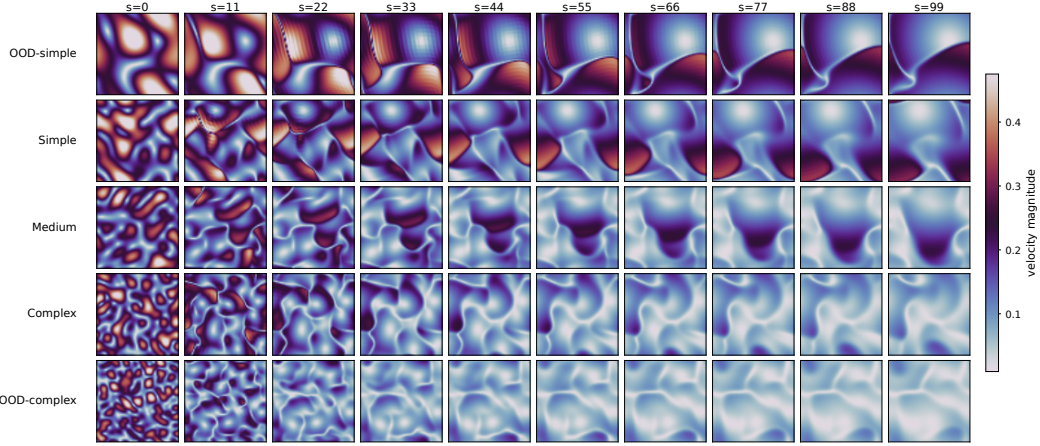


Figure 14: Burgers examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

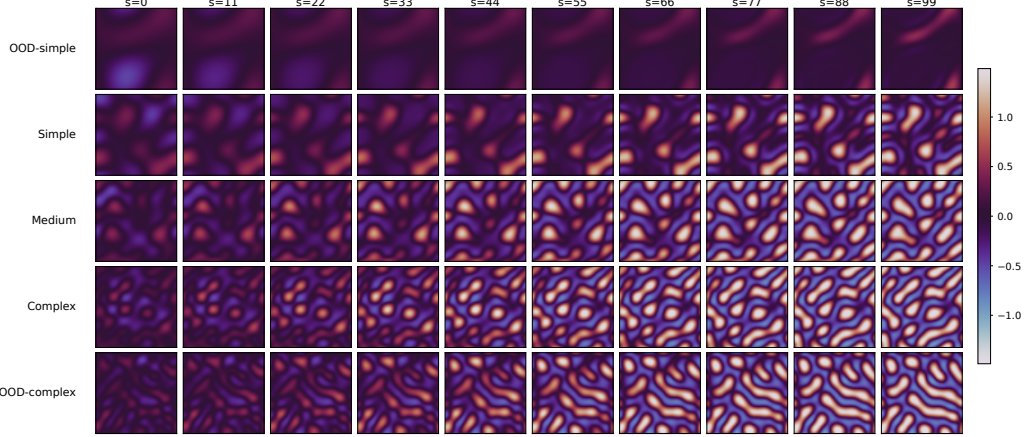


Figure 15: Swift–Hohenberg examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

amplitude 0.5, ETDRK order 2, and dealiasing fraction 1/2. Initial conditions are random truncated Fourier fields with cutoff values 1, 2, 3, 4, and 5 from OOD-simple to OOD-complex. Figure 15 shows example temporal evolution.

A.6 DECAY

The decay benchmark uses the two-dimensional incompressible Navier–Stokes equations in streamfunction–vorticity form. The stored scalar state is vorticity ω , governed by

$$\partial_t \omega + b u \cdot \nabla \omega = \lambda \omega + \nu \Delta \omega, \quad u = \nabla^\perp \psi, \quad \Delta \psi = \omega. \quad (8)$$

We use $L = 5.0$, 64×64 resolution, solver step $\Delta t = 0.06$, five internal steps per stored frame, diffusivity $\nu = 9 \times 10^{-4}$, convection scale $b = 1.0$, drag $\lambda = 0.0$, and amplitude 1.0. Initial conditions are random truncated Fourier vorticity fields with cutoff values 1, 2, 3, 4, and 5 from OOD-simple to OOD-complex. Figure 16 visualizes the vorticity field.

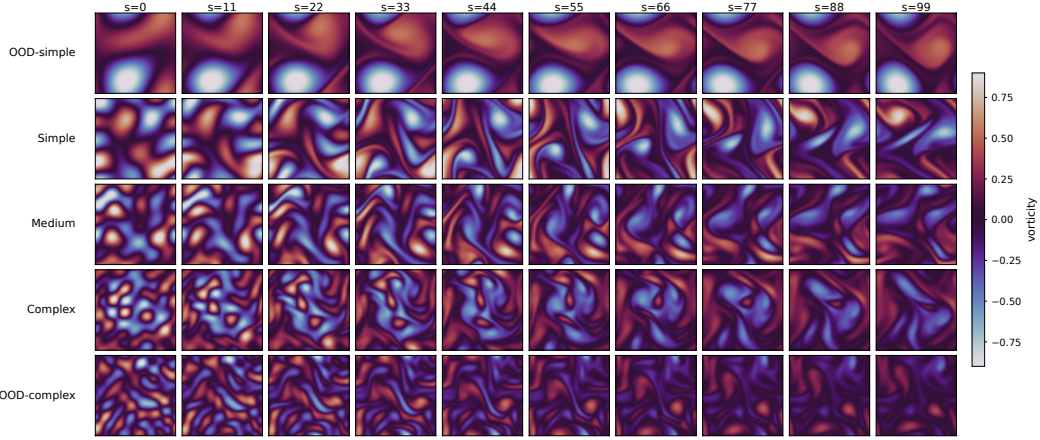


Figure 16: Navier–Stokes decay examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

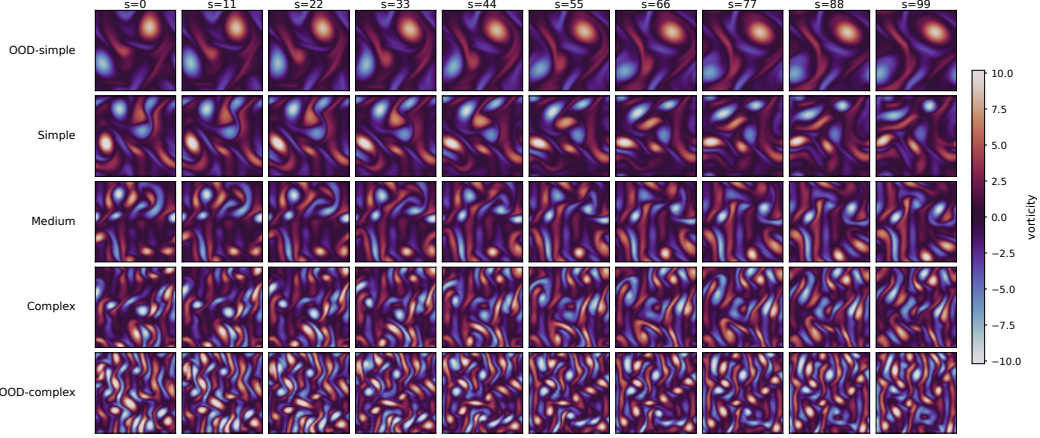


Figure 17: Kolmogorov-flow examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

A.7 KOLMOGOROV FLOW

Kolmogorov flow is also generated in streamfunction–vorticity form, but with a sinusoidal forcing term:

$$\partial_t \omega + b u \cdot \nabla \omega = \lambda \omega + \nu \Delta \omega + f_m, \quad u = \nabla^\perp \psi, \quad \Delta \psi = \omega. \quad (9)$$

Here f_m is the curl of a Kolmogorov forcing at injection mode m . We use $L = 6.0$, 64×64 resolution, solver step $\Delta t = 0.01$, two internal steps per stored frame, diffusivity $\nu = 0.01$, convection scale $b = 1.2$, drag $\lambda = -0.1$, injection scale 1.2, and amplitude 1.0. Each trajectory is simulated for 600 frames and the first 500 frames are discarded, so the stored 100 frames start after the initial transient. Complexity is controlled by the injection mode: $m = 2, 3, 4, 5, 6$ for OOD-simple, simple, medium, complex, and OOD-complex respectively. Figure 17 shows the retained vorticity dynamics.

A.8 KURAMOTO–SIVASHINSKY

We use the non-conservative, combustion-format Kuramoto–Sivashinsky equation implemented in Exponax:

$$\partial_t u + \frac{b_2}{2} \|\nabla u\|_2^2 + \psi_1 \Delta u + \psi_2 \mathcal{D}_4 u = 0, \quad (10)$$

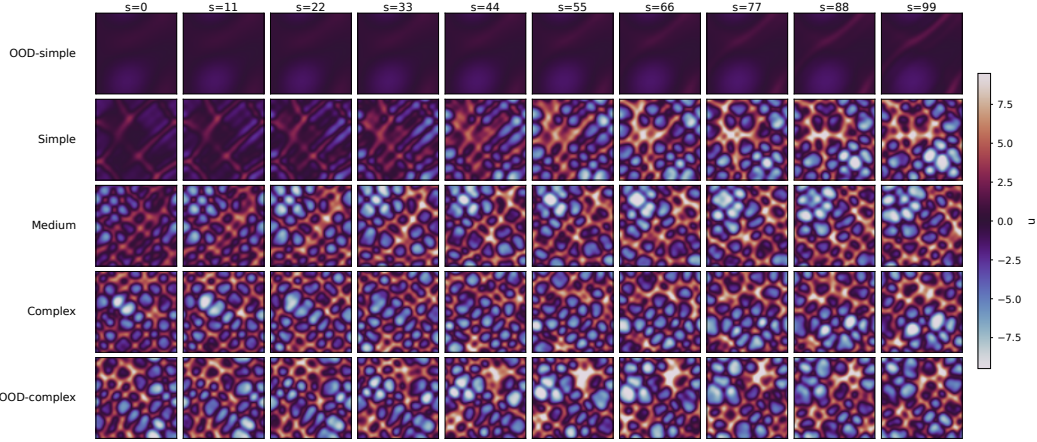


Figure 18: Kuramoto–Sivashinsky examples. Rows are IC families ordered from OOD-simple to OOD-complex; columns are uniformly sampled times.

where $\mathcal{D}_4$ denotes the Exponax fourth-order derivative operator. The benchmark uses one scalar channel, $L = 42.0$, 64×64 resolution, solver step $\Delta t = 0.1$, one internal step per stored frame, and scales $b_2 = \psi_1 = \psi_2 = 1.0$. Each trajectory is generated for 200 frames and the first 100 are discarded before saving the final 100 frames. Initial conditions are random truncated Fourier fields with cutoff values 1, 2, 3, 4, and 5 from OOD-simple to OOD-complex. Figure 18 shows examples after transient trimming.

B APPENDIX: SUPPLEMENTARY RESULTS

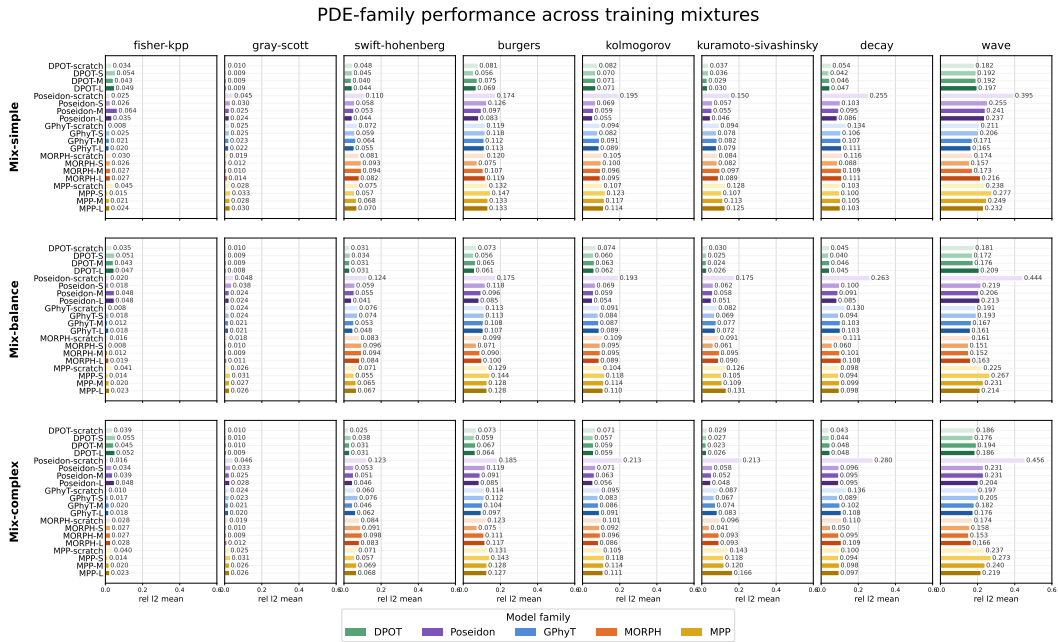


Figure 19: PDE-family raw error across training mixtures at the first predicted frame. Rows correspond to training mixtures and columns correspond to PDE families. Within each panel, bars show the raw relative L_2 error of each model variant, averaged over all 25 test regimes. Colors group model variants by model family.

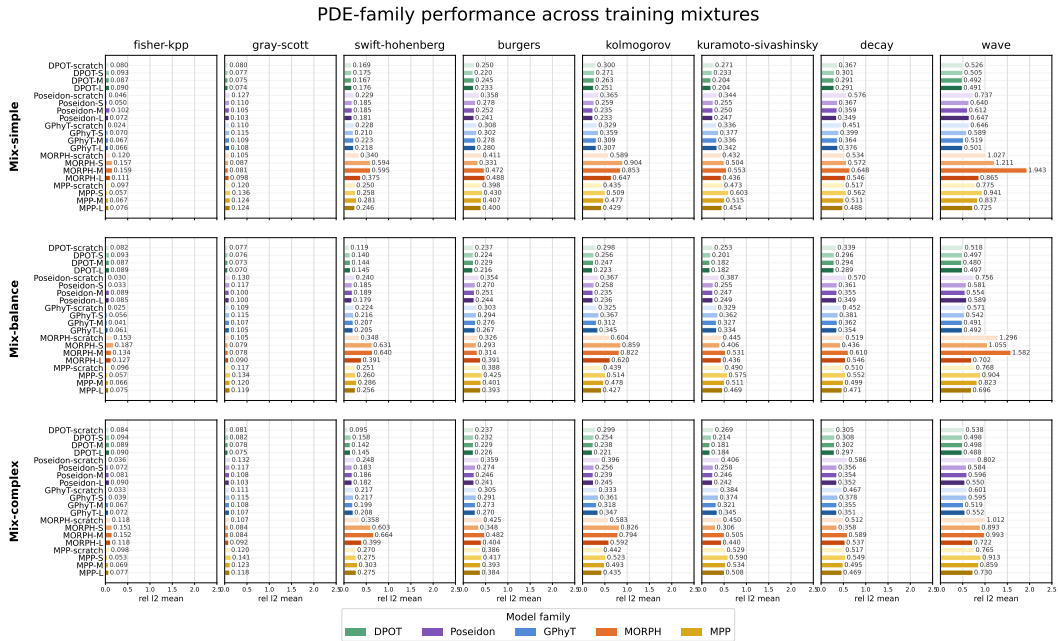


Figure 20: PDE-family raw error across training mixtures at the 10-step horizon. Rows correspond to training mixtures and columns correspond to PDE families. Within each panel, bars show the raw relative L_2 error of each model variant, averaged over all 25 test regimes and the 10 in-horizon predicted frames. Colors group variants by model family.

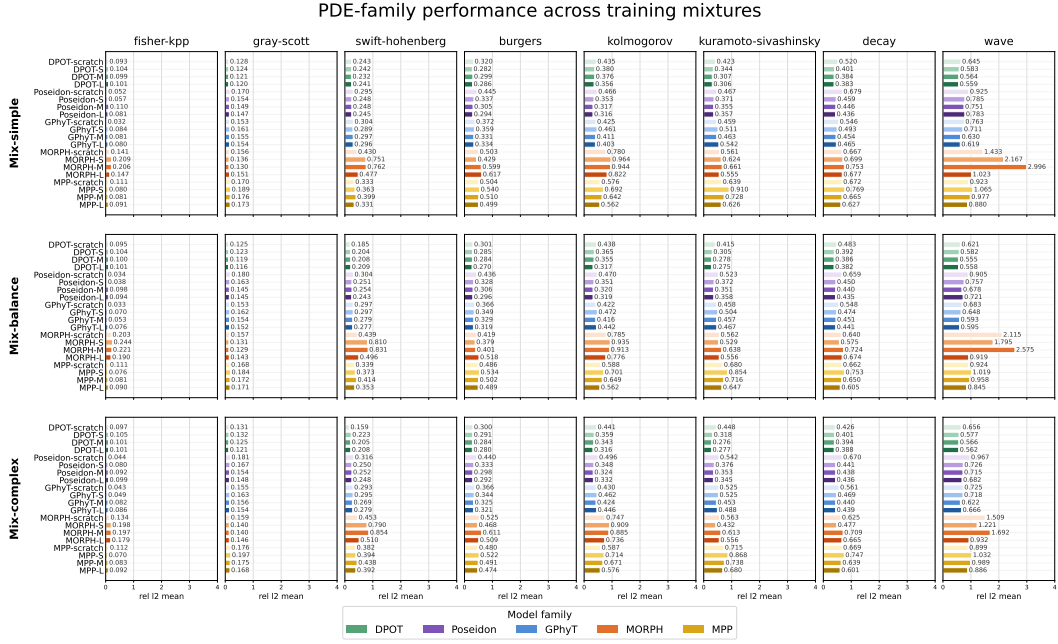


Figure 21: PDE-family raw error across training mixtures over the full 15-frame rollout. Rows correspond to training mixtures and columns correspond to PDE families. Within each panel, bars show the raw relative L_2 error of each model variant, averaged over all 25 test regimes and all 15 predicted frames. Colors group variants by model family.

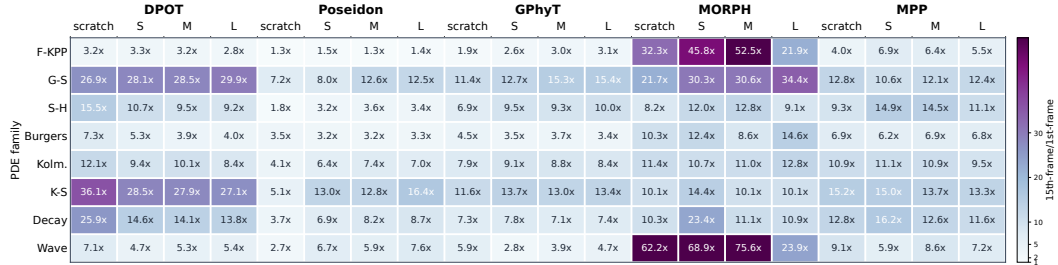


Figure 22: 15th-frame rollout amplification by model variant and PDE family. Each cell reports $E_{15\text{th-frame}}/E_{1\text{st-frame}}$, where $E_{15\text{th-frame}}$ is the error at prediction frame 15 and $E_{1\text{st-frame}}$ is the first predicted-frame error. This figure is the rollout-horizon counterpart of Figure 5.

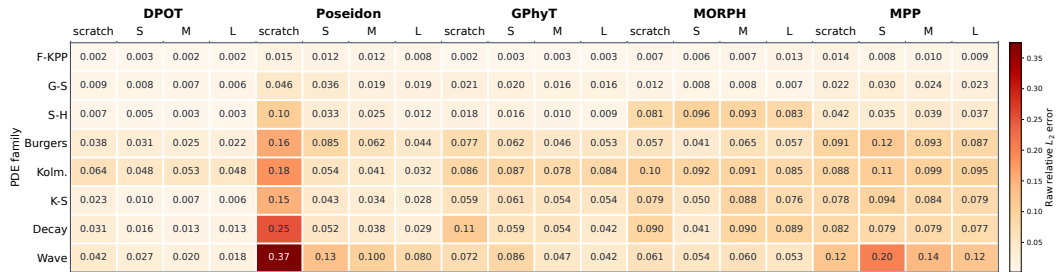


Figure 23: Raw first-frame error by model variant and PDE family. Each cell reports the raw relative L_2 error at prediction frame 1, averaged over the three train-seen cells under Mix-balance. Rows correspond to PDE families and columns correspond to model variants. The OrRd color range is normalized using the minimum and maximum values within this figure.

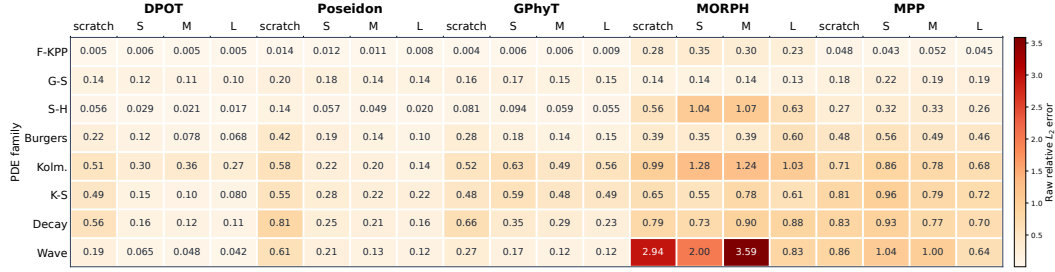


Figure 24: **Raw 10th-frame error by model variant and PDE family.** Each cell reports the raw relative L_2 error at prediction frame 10, averaged over the three train-seen cells under Mix-balance. Rows correspond to PDE families and columns correspond to model variants. The OrRd color range is normalized using the minimum and maximum values within this figure.

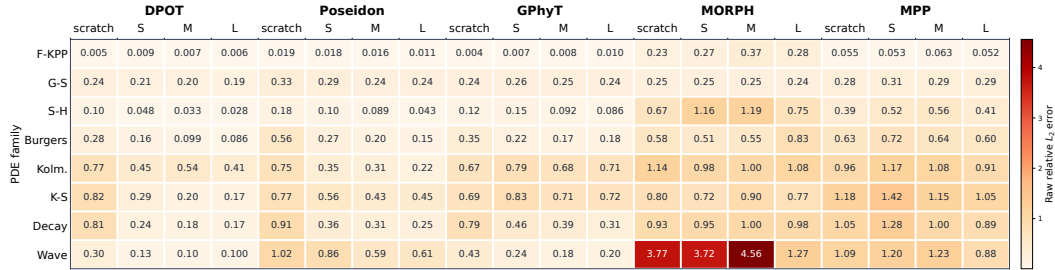


Figure 25: **Raw 15th-frame error by model variant and PDE family.** Each cell reports the raw relative L_2 error at prediction frame 15, averaged over the three train-seen cells under Mix-balance. Rows correspond to PDE families and columns correspond to model variants. The OrRd color range is normalized using the minimum and maximum values within this figure.

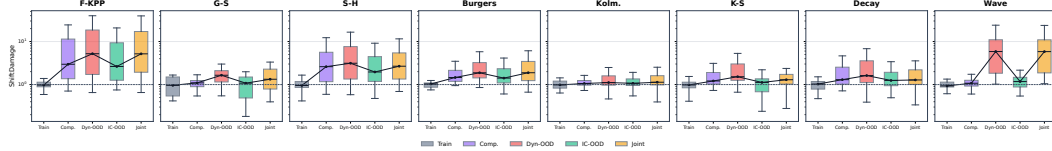


Figure 26: **ShiftDamage by shift group.** The same 10-step grouping shown separately for each PDE family. The dashed line marks no degradation relative to the train-seen baseline. Dynamic-OOD and Joint-OOD have the heaviest tails, indicating that dynamic-scale extrapolation produces larger relative robustness gaps than IC-only shifts on average, although the severity is PDE-dependent.



Figure 27: **DPOT variant-level 5×5 ShiftDamage grids.** Rows correspond to scratch/S/M/L variants, and columns correspond to PDE families. Each mini-grid shows 10-step ShiftDamage over dynamic strength and initial-condition complexity under Mix-balance. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.



Figure 28: **Poseidon variant-level 5×5 ShiftDamage grids.** Rows correspond to scratch/S/M/L variants, and columns correspond to PDE families. Each mini-grid shows 10-step ShiftDamage over dynamic strength and initial-condition complexity under Mix-balance. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.



Figure 29: **GPhyT variant-level 5×5 ShiftDamage grids.** Rows correspond to scratch/S/M/L variants, and columns correspond to PDE families. Each mini-grid shows 10-step ShiftDamage over dynamic strength and initial-condition complexity under Mix-balance. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.



Figure 30: **MORPH variant-level 5×5 ShiftDamage grids.** Rows correspond to scratch/S/M/L variants, and columns correspond to PDE families. Each mini-grid shows 10-step ShiftDamage over dynamic strength and initial-condition complexity under Mix-balance. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.



Figure 31: **MPP variant-level 5×5 ShiftDamage grids.** Rows correspond to scratch/S/M/L variants, and columns correspond to PDE families. Each mini-grid shows 10-step ShiftDamage over dynamic strength and initial-condition complexity under Mix-balance. Colors use $\log_2(\text{ShiftDamage})$: $1\times$ maps to zero/white, values below $1\times$ are green, and values above $1\times$ are red.